

The Herglotz function at rational arguments: a general closed form for $F(2/q)$, and the elementary boundary

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Source: D. Radchenko and D. Zagier, *Arithmetic properties of the Herglotz function*
(`src/PolyLog/docs/pdfs/HerglotzFunction/`). Reproducer:
`src/PolyLog/tools/scratch/herglotz/rational_harvest.py`.

Abstract

Radchenko and Zagier prove (their Theorem 3) that for every positive rational x the value $F(x) - F(1)$ of the Herglotz function is a rational combination of dilogarithms at cyclotomic arguments, and they write out two explicit cases: $F(n)$ (a pure combination of logarithms, eq. 23) and the single fractional example $F(2/5)$ (eq. 24). We turn the second into a general theorem. For *every* $q \geq 3$,

$$F(2/q) - F(1) = -\frac{(q-1)(q-2)}{12q} \pi^2 + \log^2 2 - 2 \sum_{k=1}^{\lfloor (q-1)/2 \rfloor} \log^2 \left(2 \sin \frac{\pi k}{q} \right) - \frac{1}{2} \sum_{k=1}^{\lfloor (q-1)/2 \rfloor} \text{Li}_2 \left(-\cot^2 \frac{\pi k}{q} \right),$$

with an explicit boundary correction for even q . We give a complete proof, and the companion pure-logarithm formula for $F(1/q)$. The new ingredient $\text{Li}_2(-\cot^2 \frac{\pi k}{q})$ is a dilogarithm at an algebraic argument of degree $\varphi(q)/2$: for $q = 5$ it collapses to the rational $\text{Li}_2(4/5)$ of eq. 24, but for $q \geq 7$ it does not, which is exactly why no simpler form than Theorem 3 exists there. Every statement is verified to $> 10^{-99}$ against the exact dilogarithm sum and cross-checked against the defining integral.

1 The exact dilogarithm sum

Radchenko–Zagier’s Theorem 3 is proved from the integral representation $F(P/Q) - F(1) = \int_0^1 \left(\frac{Q}{1-t^Q} - \frac{P}{1-t^P} \right) \log(1-t^P) \frac{dt}{t}$ via the distribution relations, giving the *exact, finite* evaluation

$$F\left(\frac{P}{Q}\right) - F(1) = \sum_{\alpha^P=1} \left(\sum_{\substack{\beta^Q=1 \\ \beta \neq 1}} f(\alpha, \beta) - \sum_{\substack{\beta^P=1 \\ \beta \neq 1}} f(\alpha, \beta) \right), \quad f(\alpha, \beta) = \text{Li}_2\left(\frac{\beta}{\beta-1}\right) - \text{Li}_2\left(\frac{\alpha-\beta}{1-\beta}\right). \quad (1)$$

Because (??) is a finite sum of dilogarithms, it evaluates $F(P/Q) - F(1)$ to arbitrary precision with no quadrature error; we use it as ground truth, cross-checked against the defining integral.

2 Two reductions on the line $\operatorname{Re} = \frac{1}{2}$

The whole calculation rests on one elementary observation. For a unit $\beta = e^{i\theta}$,

$$\frac{\beta}{\beta-1} = \frac{1}{1-\beta^{-1}} \text{ has real part } \frac{1}{2}, \quad \arg \frac{\beta}{\beta-1} = \frac{\theta}{2} - \frac{\pi}{2}, \quad \left| \frac{\beta}{\beta-1} \right| = \frac{1}{2 \sin(\theta/2)}. \quad (2)$$

By the reflection $\operatorname{Li}_2(z) + \operatorname{Li}_2(1-z) = \frac{\pi^2}{6} - \log z \log(1-z)$ with $1-z = \bar{z}$ on the line $\operatorname{Re} z = \frac{1}{2}$,

$$2 \operatorname{Re} \operatorname{Li}_2\left(\frac{1}{2} + iy\right) = \frac{\pi^2}{6} - \log^2 |z| - (\arg z)^2. \quad (3)$$

The second reduction handles the $\alpha = -1$ part of (??): for $\beta = e^{i\theta}$, $\frac{-1-\beta}{1-\beta} = -i \cot \frac{\theta}{2}$ is purely imaginary, and for real y

$$\operatorname{Re} \operatorname{Li}_2(iy) = \frac{1}{4} \operatorname{Li}_2(-y^2), \quad (4)$$

from $\operatorname{Li}_2(iy) + \operatorname{Li}_2(-iy) = \frac{1}{2} \operatorname{Li}_2(-y^2)$ together with $\operatorname{Li}_2(-iy) = \overline{\operatorname{Li}_2(iy)}$.

3 $F(1/q)$: pure logarithms

Theorem 1. For all $q \geq 2$, with $m = \lfloor (q-1)/2 \rfloor$,

$$F\left(\frac{1}{q}\right) - F(1) = -\frac{(q-1)(3q-2)}{24q} \pi^2 - \sum_{k=1}^m \log^2\left(2 \sin \frac{\pi k}{q}\right) - [q \text{ even}] \frac{1}{2} \log^2 2.$$

Proof. With $P = 1$ only $\alpha = 1$ occurs and the inner β^P -sum is empty, so (??) reads $F(1/q) - F(1) = \sum_{\beta^q=1, \beta \neq 1} (\operatorname{Li}_2(\frac{\beta}{\beta-1}) - \operatorname{Li}_2(1))$. Pairing each $\beta = e^{2\pi i k/q}$ with $\bar{\beta}$ and applying (??) with $|z| = 1/(2 \sin \frac{\pi k}{q})$ and $\arg z = \frac{\pi k}{q} - \frac{\pi}{2}$ gives, after subtracting $(q-1)\frac{\pi^2}{6}$, the formula with the π^2 coefficient $-\left[\frac{q-1}{12} + \sum_{k=1}^m (\frac{k}{q} - \frac{1}{2})^2\right]$, which equals the clean rational $-\frac{(q-1)(3q-2)}{24q}$. For even q the self-conjugate root $\beta = -1$ contributes $\operatorname{Li}_2(\frac{1}{2}) = \frac{\pi^2}{12} - \frac{1}{2} \log^2 2$, supplying the extra $-\frac{1}{2} \log^2 2$. $\square$

This is the conjecturally complete list (with $F(n)$, eq. 23) of pure-logarithm values: only $x = n$ and $x = 1/n$ avoid genuine dilogarithms.

4 $F(2/q)$: the general theorem

Theorem 2. For every $q \geq 3$,

$$F\left(\frac{2}{q}\right) - F(1) = -\frac{(q-1)(q-2)}{12q} \pi^2 + \log^2 2 - 2 \sum_{k=1}^{\lfloor (q-1)/2 \rfloor} \log^2\left(2 \sin \frac{\pi k}{q}\right) - \frac{1}{2} \sum_{k=1}^{\lfloor (q-1)/2 \rfloor} \operatorname{Li}_2\left(-\cot^2 \frac{\pi k}{q}\right),$$

where for even q the two sums run to $k = \frac{q}{2} - 1$ and a further $-\log^2 2$ is added.

Proof (odd q). Here $P = 2$, so $\alpha \in \{1, -1\}$ and the inner β^P -sum is the single term $\beta = -1$. Split (??) as $(\sum_{\alpha} \sum_{\beta^q \neq 1} f(\alpha, \beta)) - \sum_{\alpha} f(\alpha, -1)$. For the first part, $f(1, \beta) + f(-1, \beta) = 2 \operatorname{Li}_2(\frac{\beta}{\beta-1}) - \frac{\pi^2}{6} - \operatorname{Li}_2(\frac{-1-\beta}{1-\beta})$. Summing over $\beta \neq 1$: the $\operatorname{Li}_2(\frac{\beta}{\beta-1})$ pieces give $2S_1$ with $S_1 = \sum_{k=1}^{(q-1)/2} [\frac{\pi^2}{6} - \log^2(2 \sin \frac{\pi k}{q}) - \pi^2(\frac{k}{q} - \frac{1}{2})^2]$ by (??); the constants give $-(q-1)\frac{\pi^2}{6}$; and by (??) (with $\frac{-1-\beta}{1-\beta} =$

$-i \cot \frac{\pi k}{q}$, pairing $k \leftrightarrow q - k$) the last pieces give $-\frac{1}{2} \sum_{k=1}^{(q-1)/2} \text{Li}_2(-\cot^2 \frac{\pi k}{q})$. For the boundary, $\sum_{\alpha} f(\alpha, -1) = 2 \text{Li}_2(\frac{1}{2}) - \frac{\pi^2}{6} = -\log^2 2$ using $\text{Li}_2(\frac{1}{2}) = \frac{\pi^2}{12} - \frac{1}{2} \log^2 2$, so $-\sum_{\alpha} f(\alpha, -1) = +\log^2 2$. Collecting,

$$F(2/q) - F(1) = 2S_1 - (q-1)\frac{\pi^2}{6} - \frac{1}{2} \sum \text{Li}_2(-\cot^2 \frac{\pi k}{q}) + \log^2 2,$$

and the $\frac{\pi^2}{6}$ terms of $2S_1$ cancel the $-(q-1)\frac{\pi^2}{6}$, leaving the π^2 coefficient $-2 \sum_{k=1}^{(q-1)/2} (\frac{k}{q} - \frac{1}{2})^2$. The elementary identity $2 \sum_{k=1}^{(q-1)/2} (\frac{k}{q} - \frac{1}{2})^2 = \frac{(q-1)(q-2)}{12q}$ finishes the proof. The even- q case is identical except that $\beta = -1$ is now a genuine q -th root, contributing the displayed boundary term. $\square$

The coefficient is exactly twice that of eq. 23 for $F(q)$. For $q = 5$ the two terms $\text{Li}_2(-\cot^2 \frac{\pi}{5}) + \text{Li}_2(-\cot^2 \frac{2\pi}{5})$ collapse (the real subfield is quadratic) to the single $\text{Li}_2(4/5)$ of eq. 24; Theorem ?? is the general statement behind that one example.

5 The elementary boundary for $F(p/q)$

The collapse in Theorem ?? is special to $p = 2$, where the $\alpha = -1$ arguments lie on the imaginary axis (??). For $p \geq 3$ the arguments $(\alpha - \beta)/(1 - \beta)$ are cyclotomic ratios $(1 - \zeta_p^a)/(1 - \zeta_q^b)$ lying off both special lines, and no $-\cot^2$ form exists. Numerically (PSLQ with a Champernowne canary, 130 digits), $F(p/q) - F(1)$ reduces to dilogarithms at *rational* arguments exactly when the maximal real subfield $\mathbb{Q}(\zeta_q)^+$ has degree ≤ 2 — i.e. $q \in \{3, 4, 5, 6, 8, 12\}$ — and otherwise needs genuine Li_2 at arguments of degree $\varphi(q)/2$. The harvested table of collapsing cases is in `src/PolyLog/tools/scratch/herglotz/harvest_table.py`.

Proposition 3 (Collapse criterion). *For coprime p/q , the value $F(p/q) - F(1)$ reduces to dilogarithms at rational arguments (together with π^2 and logarithms) if and only if $\text{rad}(q) \mid 30$, i.e. every prime factor of q lies in $\{2, 3, 5\}$; otherwise it requires genuine Li_2 at the cyclotomic-ratio arguments. Across all such rational cases only two new dilogarithm constants ever occur: $\text{Li}_2(1/3)$ (conductor 3) and the pair $\{\text{Li}_2(1/5), \text{Li}_2(2/5)\}$ (conductor 5).*

This was found by PSLQ with a Champernowne canary and re-verified against the exact sum (??); e.g. $F(3/4) - F(1) = -\frac{19}{144}\pi^2 - \frac{3}{4}\log^2 2 - 2\log 2 \log 3 + \frac{7}{4}\log^2 3 + 2\text{Li}_2(\frac{1}{3})$ and $F(4/5) - F(1) = \frac{3}{80}\pi^2 + \frac{11}{4}\log^2 2 - \frac{1}{2}\log^2 5 - \text{Li}_2(\frac{1}{5}) - \log^2(2 \sin \frac{\pi}{5}) - \log^2(2 \sin \frac{2\pi}{5})$.

6 Further general laws

The same machinery yields three more closed-form families and a derivative law; all are verified to $> 10^{-149}$ against (??) (and cross-checked against the integral) by an independent multi-agent pass.

The $F(n/(n+1))$ family (Radchenko–Zagier’s Example 2, made explicit). For all $n \geq 2$,

$$F\left(\frac{n}{n+1}\right) - F(1) = -\frac{3n^2 + 3n + 2}{24n(n+1)}\pi^2 + \text{Li}_2\left(\frac{n}{n+1}\right) + \frac{1}{2}\log^2 \frac{n+1}{n} - \frac{1}{2}S(n+1) + \frac{1}{2}S(n), \quad S(q) := \sum_{k=1}^{q-1} \log^2\left(2 \sin \frac{\pi k}{q}\right),$$

the single dilogarithm $\text{Li}_2(n/(n+1))$ appearing with coefficient exactly $+1$ (the paper’s bilinear-log claim).

The reciprocity law for $F(p/2)$. For all odd $p \geq 3$,

$$F\left(\frac{p}{2}\right) + F\left(\frac{2}{p}\right) - 2F(1) = -\frac{(p-2)^2}{12p}\pi^2 + \frac{1}{2}\log^2 \frac{p}{2}.$$

The $\text{Li}_2(-\cot^2 \frac{\pi k}{p})$ terms of $F(p/2)$ are exactly minus those of $F(2/p)$ and cancel in the sum; combined with Theorem ?? this gives $F(p/2)$ in closed form.

The derivative law $F'(2/q)$ (Radchenko–Zagier Proposition 4). For all $q \geq 3$, writing $T(q) := \frac{2}{q}F'(2/q) - (1 + \log 2)$,

$$T(q) = \frac{q^2 - 4}{24q}\pi^2 + W(q), \quad W(q) = \sum_{k=1}^{q-1} \cot \frac{2\pi k}{q} \text{Cl}_2\left(\frac{2\pi k}{q}\right),$$

with an extra $-\log 2$ for even q . Unlike $F(2/q)$ itself, this does *not* reduce to elementary form: $W(q)$ is a genuine Clausen combination, irreducible over the vocabulary of Theorem ??.

The values $J(p/q)$. With $J(x) = F(2x) - 2F(x) + F(x/2) + \frac{\pi^2}{12x}$, the combination $J(2/q)$ is a *pure-logarithm* value if and only if $q \in \{3, 5\}$ — so Radchenko–Zagier’s stated $J(2/5) = \frac{11\pi^2}{240} + \frac{3}{4}\log^2 2 - 2\log^2 \frac{\sqrt{5}+1}{2}$ (their only worked J example) is a special case, not a general law (the dilogarithms cancel only at $q = 3, 5$); we also record $J(2/3) = -\frac{\pi^2}{144} + \frac{3}{4}\log^2 2$. The values $J(1/q)$ always retain the $\text{Li}_2(-\cot^2)$ dilogarithms inherited from $F(2/q)$.

Verification and reproducibility

The exact sum (??), the formulas of Theorems ??–?? (verified for $q = 3, \dots, 40$ to better than 10^{-99} , odd and even, prime and composite), the step-by-step audit of the reductions (??)–(??), and the collapse table are in `src/PolyLog/tools/scratch/herglotz/` and recorded in `src/PolyLog/identities/herglotz-rational-values.wl`.